Parent's signature

Name	Index Number	CTG

YISHUN JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2018

BIOLOGY **8876/02**

HIGHER 1 **29 AUGUST 2018**

Paper 2 Structured and Free-response Questions **Wed 1400 - 1600**

2 hours

Candidates answer on the Question Paper.
No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name and CTG in the spaces at the top of this page and on all separate answer paper used.

Write in dark blue or black pen only.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** questions in the spaces provided on the Question paper.

Section B
Write your answers in the spaces provided on the Question Paper.

At the end of the examination, fasten all your work securely together.

This question paper consists of **20** printed pages.

FOR EXAMINER'S USE	
Paper 1	/ 30
Paper 2	
Section A	
Q1	/14
Q2	/10
Q3	/ 6
Q4	/ 5
Q5	/ 10
Section B	
Q6 OR 7	/ 15
Total	/ 60
Overall	/ 90

Section A

Answer **all** questions in this section.

1 In an experiment to study the effect of heat treatment on the digestibility of protein substrate and the effect of raw bean extract on protease activity, various reaction mixtures were prepared and were incubated for 30 minutes.

The protein concentration of each reaction mixture at the beginning and at the end of the incubation period was determined by the colorimetric method which measures colour intensity of these reaction mixtures. The results were shown in Table 1.1.

Table 1.1

Incubation period /min	**Colour intensity of the reaction mixture / arbitrary unit**		
	Tube A	**Tube B**	**Tube C**
	Protease + heated protein substrate	**Protease + unheated protein substrate**	**Protease + unheated protein substrate + raw bean extract**
0	10	10	10
30	4	7	9

Fig. 1.1 shows a standard graph obtained by using colorimetric method for determining concentration of protein solutions.

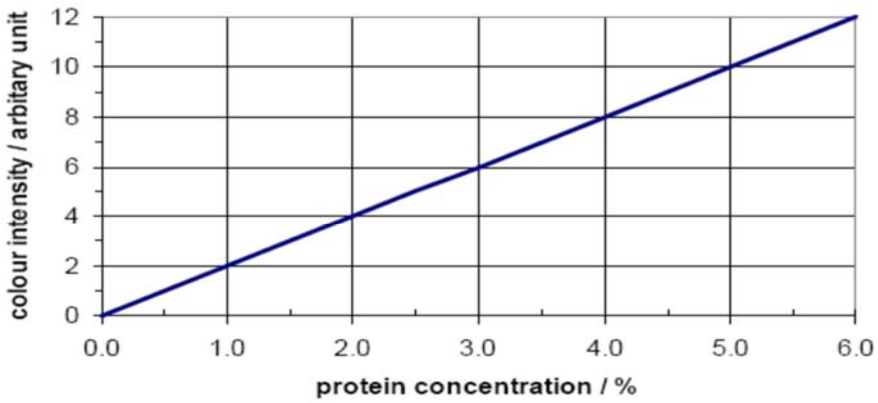

Fig. 1.1

(a) With reference to Table 1.1 and Fig 1.1,

(i) Determine the protein concentration of the three tubes at time 0 min.

5% protein concentration [1]

(ii) Explain which protein substrate is more digestible by protease.

1. heated protein more digestible by protease;
2. less protein remained after incubation with protease / lowest colour intensity
3. unheated vs heated = 3% decreases [2]

(b) Based on your knowledge of protein structure, explain the effect of heat treatment on the digestibility of this protein substrate.

1. heating increases the digestibility of this protein substrate / digest more quickly / easier to digest;
2. heating causes protein substrate denaturation/ loss of 3D configuration;
3. breaking of bonds – H, hydrophobic, ionic, disulphide (name at least 2);
4. increase in surface area of substrate for protease action/ easier access of enzyme(s' active site) to substrate/peptide bonds; [3]

(c) Suggest the effect of raw bean extract on protease activity.

inhibits protease activity [1]

(d) Dextrin is a generic term applied to a variety of products obtained by heating starch in the presence of small amounts of moisture and an acid. Dextrins can be made from starch and is a polysaccharide similar to amylopectin.

Describe the differences in molecular structure between cellulose and dextrins.

1. **Cellulose** is a polymer of around 10,000 **β-glucose molecules while d**extrin is a polymer of **α glucose molecules**,
2. Dextrin joined by **α (1→4) and α (1→6) glycosidic bonds at the branch points** while **cellulose** joined by **β (1→4) glycosidic bonds**,
3. In cellulose, **alternate β glucose molecules rotate 180^0** while α glucose molecules of dextrin are at the **same orientation** (i.e. do not rotate)
4. Dextrin is a **branched** chain of α glucose units, and **coiled into a helix while cellulose chains are long unbranched chains with a**djacent cellulose chains form hydrogen bonds / cross bridges between the chains to form **microfibrils.** [2]

Fig. 1.2 shows the structure of a triglyceride. Fig. 1.3 shows the structure of a glycolipid.

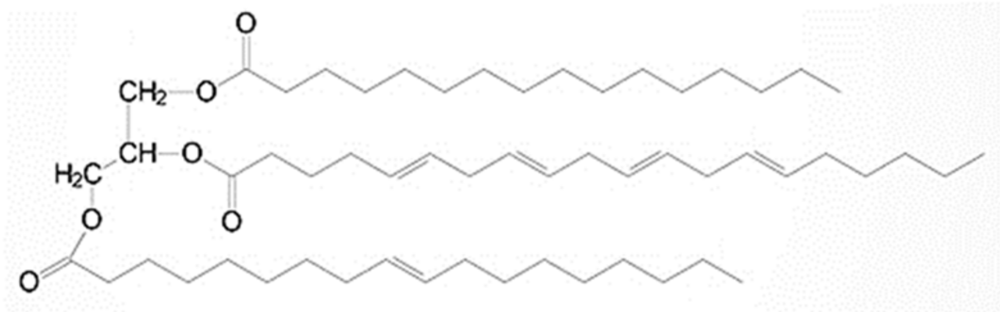

Fig. 1.2

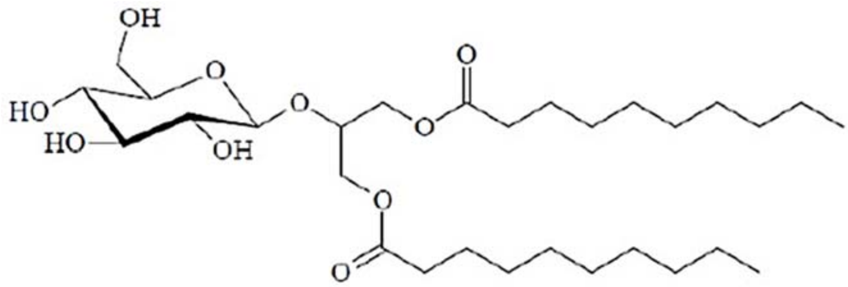

Fig. 1.3

(e) Explain the difference in solubility between triglycerides and the products of their hydrolysis.

1. Triglycerides are **non-polar molecules** that are **insoluble in water**. This is due to their incapability of forming hydrogen bonds with water molecules.
2. **Glycerol and fatty acids** are the **products of triglyceride hydrolysis**.
3. **Glycerol is an alcohol** that is **soluble** due to its ability (-OH groups in glycerol) to form hydrogen bonds with water, but **fatty acids** are **insoluble** due to their **long hydrocarbon tails** that are **non-polar** and **hydrophobic.** [3]

(f) Describe the roles of triglycerides and glycolipids in relation to the differences in their molecular structures.

1. **Has long hydrocarbon chains, acting as energy store that yield more energy** (38kJ per gram) than same mass of carbohydrates (17 kJ per g)
2. Releases **more metabolic water** during fat oxidation/ respiration.
3. Lipid molecule with a carbohydrate component and act as a recognition site for cell adhesion [2]

[Total: 14]

2 Fig. 2.1 shows two pairs of homologous chromosome carrying alleles **F/f and G/g** respectively.

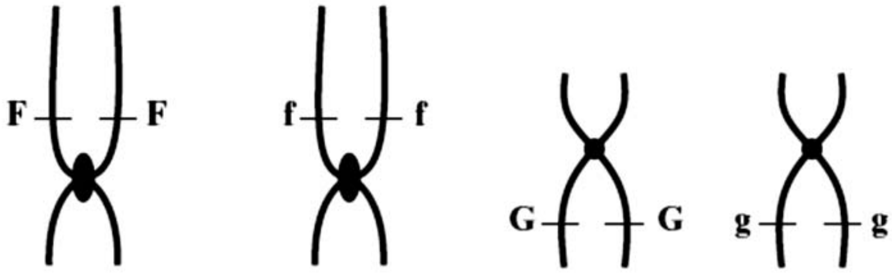

Fig. 2.1

(a) State the genotype of this cell.

FfGg [1]

(b) Explain why there are two copies of each allele for each gene present on the chromosomes in Fig. 2.1, before the start of nuclear division. [2]

1. DNA replication has occurred;
2. During interphase / S-phase of interphase before mitosis;
3. Giving rise to two chromatids;
4. With two copies of the allele, one on each chromatid;

Fig. 2.2 shows a molecule of tRNA and the enzyme that attaches the correct amino acid to it.

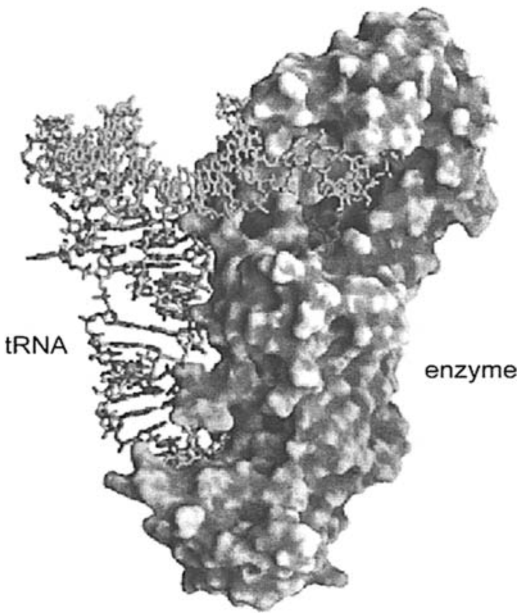

Fig. 2.2

(c) Describe **two** structural features which adapt tRNA to its role in translation.

1 3' / (CCA end) = allows <u>specific</u> amino acid molecule binds to the tRNA

2 loop with <u>anticodon</u> /region has specific triplet base sequence - <u>complementary to a codon on mRNA.</u> [2]

(d) Explain how the enzyme in Fig. 2.2 is suited to its function.

1) Identify the enzyme as aminoacyl-tRNA synthetase
2) has recognition/active site for both amino acid
3) and tRNA with specific anticodon;
4) with site being complementary in shape and charge to the substrates
5) able to catalyse formation of covalent bond between tRNA and amino acid
6) twenty different aminoacyl-tRNAase enzymes for each amino acid to ensure specificity in binding

[3]

(e) Fig. 2.3 below shows the base sequence on a length of mRNA and the amino acids represented by the respective codons.

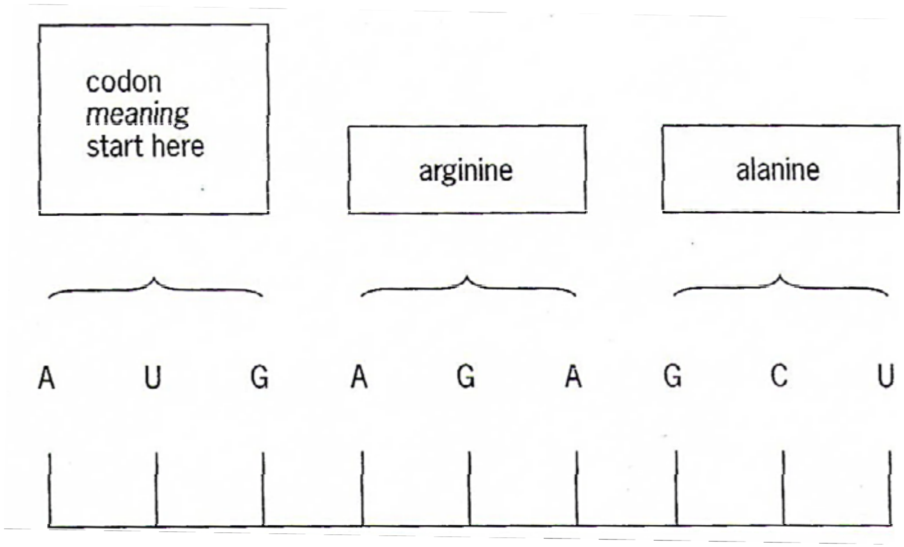

Fig. 2.3

The codon AGA codes for the amino acid arginine. However, arginine can also be represented by the codon AGG.

(i) Explain why the substitution of A by G in this case does not affect the amino acid represented.

1. Genetic code is degenerate/redundancy in genetic code;
2. 1 amino acid can be coded by more than 1 codon;
3. First two bases are most important in determining which amino acid is attached;
4. The third base is a wobble base/not as important;
5. It is a form of silent mutation [1]

(ii) The codons in the mRNA sequence do not overlap. Suggest one advantage of this.

1. Substitution of a base will only change one amino acid instead of two;;
2. Overlapping can cause difficulties in translation when the tRNA comes into the P site as it would be blocked by the previous tRNA;;
3. There is less restriction on the order of amino acids since the overlapped base of one codon has no effect on the next codon;; [1]

[Total: 10]

3 An inbred variety of maize, **A**, with finely striped leaves was found to be resistant to the fungus that causes the disease, corn leaf blight.

Plants of variety **A** were crossed with another inbred variety of maize, **B**, which had entirely green leaves and low resistance to the fungus. All the F_1 generation had entirely green leaves and low resistance.

(a) Using appropriate symbols and indicating what they represent, state the genotype of **A**.

G - entirely green leaf, R – low resistance
genotype of A: ggrr; [1]

The above F_1 generation was back-crossed to variety **A**.

(b) Draw a genetic diagram to explain the expected results of the F_2 generation from this cross.

Phenotype	***entirely green, low resistance***	***x***	***striped leaves, resistant***	***[1]***
[Genotype	***GgRr***	***x***	***ggrr***	***[2]***
Meiosis				
Gametes	***GR Gr gR gr***		***gr***	***[3]***
Fertilisation				

Using Punnett Square

	GR	***gR***	***Gr***	***gr***
gr	***GgRr***	***ggRr***	***Ggrr***	***ggrr***

F_2 genotype	***GgRr***	***ggRr***	***Ggrr***	***ggrr***	***[4]***
F_2 phenotype (offspring)	***entirely green, low resistance***	***striped leaves, low resistance***	***entirely green, resistant***	***striped leaves, resistant***	***[5]***
F_2 phenotypic ratio	***1 :***	***1 :***	***1 :***	***1***	

[5]

[Total: 6]

4 Dengue fever is an infectious disease transmitted by a vector.

(a) **(i)** State the name of the pathogen that causes dengue fever.

Dengue virus; **A** DENV [1]

(ii) State the name of the vector that transmits the pathogen.

Aedes aegypti **R** mosquito, aedes, *A. aegypti* [1]
R genus and species are not underlined separately

In 2002, the World Health Organization estimates that there might be 50 million cases of dengue infection worldwide every year. In Singapore, dengue fever is endemic with year-round transmission. Fig 4.1 shows the incidences of dengue fever from 1985 to 3 September 2005.

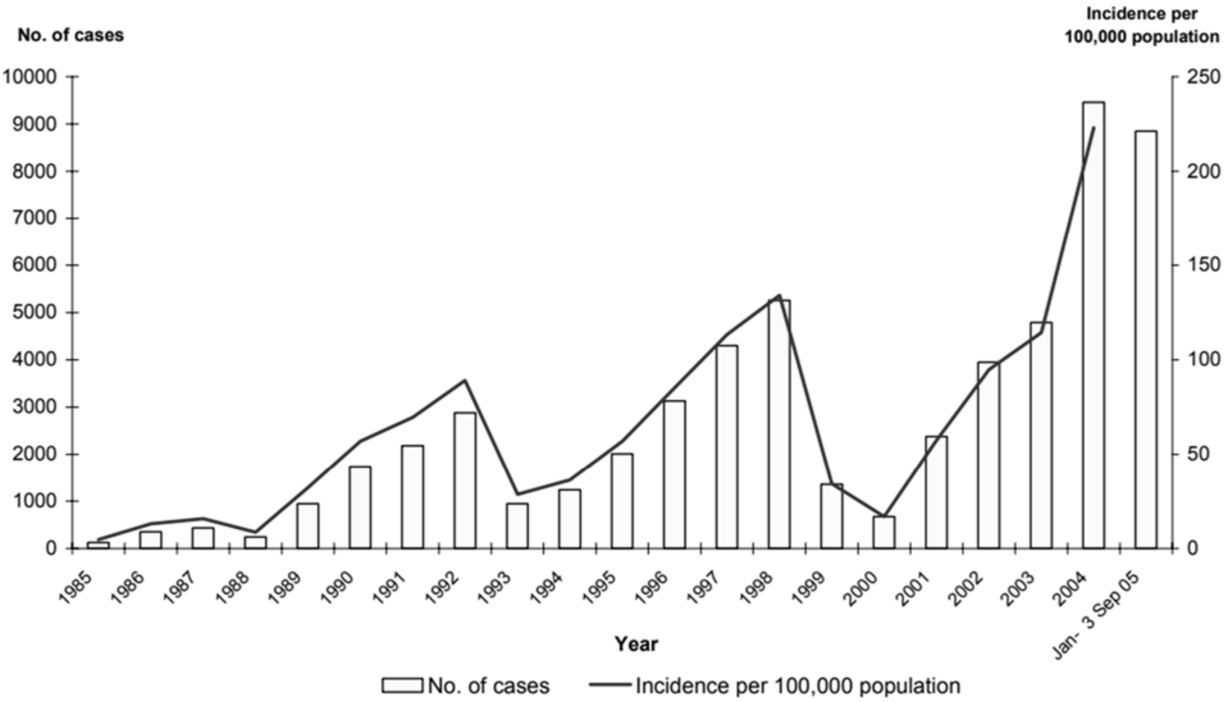

Fig. 4.1

(b) Suggest **two** reasons why the number of cases of dengue fever decreased in 1993 and 2000.

Give example of control of breeding of mosquitoes;
e.g. drainage of stagnant water / sterile males / aerial spraying of insecticide / applying oil on water / breeding of fishes in water

example of reduction of contact between mosquitoes and human;
e.g. use of bed nets / insect repellents

Improve in surveillance (by NEA);

Improve awareness of / education about, transmission / control methods;

AVP e.g. drop in ambience temperature → not suitable for breeding of mosquitoes

[Any 2] [2]

(c) Dengue fever is very difficult to control even though there has been improved understanding of the disease.

Explain why dengue fever is very difficult to control.

1 no vaccine **; A** no effective vaccine
2 any problem in developing a vaccine **;**
e.g. 4 serotypes of dengue viruses / antibody dependent enhancement of dengue virus infection
3 insecticide resistance in *Aedes* **;**
any example, e.g. DDT / dieldrin / pyrethroids **;**
4 *ref. to* conditions for breeding of *Aedes* **;**
5 problems with, funding research / AW **;**
6 cost of, drugs / insecticides, to government / health authorities / individuals **;**
7 people with HIV / AIDS are at high(er) risk than others **;**
8 lack of knowledge / lack of education / 'fatalism' / AW **;**
9 inaccessibility of some regions to healthcare **;**
10 infected people not, identified / diagnosed **;**
11 AVP **;**
e.g. migration of people with dengue fever to places without malaria; global warming promoting the growth and development of mosquitoes;

[Any 1] [1]

[Total: 5]

5 Fig. 5.1 shows the Calvin cycle which takes place inside a plant cell.

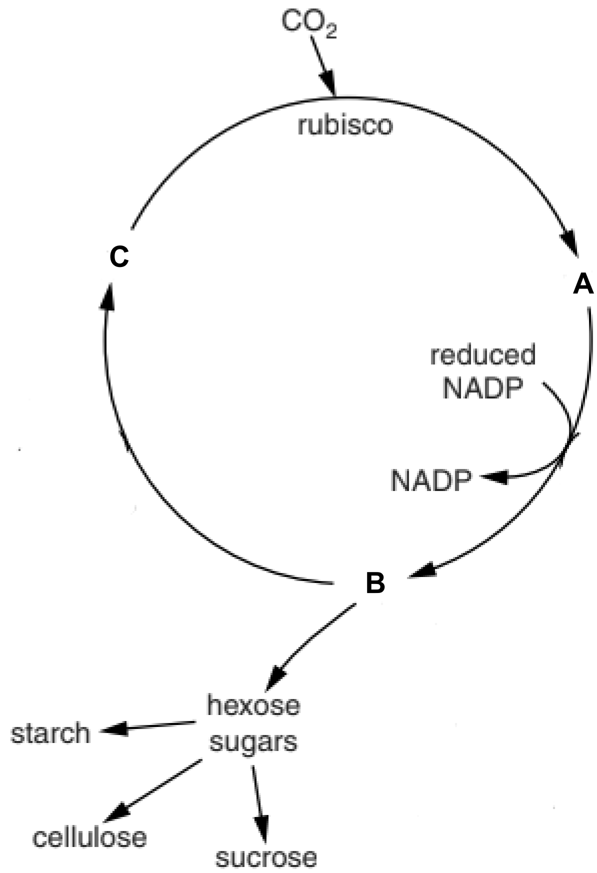

Fig. 5.1

(a) Identify substances **A**, **B** and **C**.

A – glycerate-3-phosphate / 3-phosphoglycerate (R! G3P, 3PG)

B – glyceraldehyde-3-phosphate / triose phosphate (R! G3P, TP)

C – ribulose bisphosphate / RuBP [3]

(b) Describe the role of NADP in linking the light dependent reactions to the Calvin cycle as shown in Fig. 5.1.

NADP is the final electron acceptor of the non-cyclic photophosphorylation, reduced to NADPH;

NADPH is required to reduced glycerate-3-phosphate (A! if student wrote 1,3-bisphosphoglycerate) to glyceraldehyde-3-phosphate; [2]

(c) In aerobic respiration and photosynthesis, ATP is synthesised via chemiosmosis in oxidative phosphorylation and photophosphorylation.

(i) Outline the role of stalked particles in chemiosmosis in oxidative phosphorylation.

Protons diffuse from intermembrane space back to matrix via the stalked particles, results in proton motive force;

Proton motive force drives ATP synthase to catalyse the formation of ATP from ADP and free phosphate; [2]

(ii) Explain why less ATP is synthesised from the same mass of glucose in anaerobic respiration compared to aerobic respiration.

anaerobic – accept ora for aerobic
1 idea that glucose not completely, broken down/oxidised
or
only glycolysis occurs;
or
pyruvate/lactate/ethanol, still contains energy;

2 ETC stops (because) no oxygen to act as (final) electron acceptor;

3 (so) no, Krebs cycle/link reaction/oxidative phosphorylation / chemiosmosis; [3]

[Total: 10]

Section B

Answer **one** question in this section.

Write your answers on the lined papers provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where applicable.

Your answers must be set out in sections **(a)** and **(b)**, as indicated in the question.

6 **(a)** Discuss the role of complementarity in cellular mechanisms. [6]

Complementary shape (max 2)

1. Substrate(s) fit into the active site of enzyme;
2. via lock and key hypothesis;
3. And induced fit hypothesis;
4. To form enzyme-substrate complex;

5. DNA to fit into binding site of proteins
6. To regulate replication;
7. And gene expression;

8. Binding of substances to transport proteins;
9. Allows for movement of substances across cell membrane;
10. and viral entry;

Complementary interaction (max 2)

11. H bonds between polar groups;
12. Hydrophobic interaction between non-polar groups;
13. Ionic bonds between oppositely charged groups;
14. Allows for folding of polypeptide into 3D shape;
15. Stability of biomolecules;

Complementary base pair (max 2)

16. A-T (A-U) and C-G;
17. Allows for stability of DNA double helix;
18. Allows for replication of DNA;
19. Allows for the synthesis of mRNA/transcription;
20. allows for the binding of (anticodon on) tRNA to (codon on) mRNA;

(b) Red blood cells are replaced every 3 to 4 months.

Explain how haemoglobin is synthesised when haematopoietic stem cells form red blood cells. [9]

<u>cells and organelles: max 2</u>

1. haemoglobin a- and b- genes in DNA, DNA in nucleus;
2. transcription of haemoglobin a- and b- gene in nucleus by RNA polymerase;
3. export of mature mRNAs into cytoplasm via nuclear pores;
4. free 80S ribosomes bind to each mRNA, translation of each mRNA to form a-chain and b-chain separately;
5. each chain fold into the tertiary structure of a-globin and b-globin;

<u>proteins: max 2</u>

6. haemoglobin - quaternary structure, 4 polypeptide chains; 2 a- and 2 b-;
7. maintained by 3 types of intermolecular bonds;
8. ref to the primary, secondary, tertiary structure of globin;

<u>stem cells: max 2</u>

9. Hematopioetic stem cells multipotent;
10. can differentiate into myeloid progenitor, myleoid progenitor cell give rise to RBC;
11. ref to internal and external signals to stimulate differentiation;

<u>DNA & genomics: max 2</u>

12. ref to transcription mechanism;
13. ref to translation mechanism;
14. primary RNA from transcription needs to be processed (post-transcriptional modification) to form mature mRNA;
15. removal of introns by RNA excision; splicing of exons;
16. additional of 5'-methylguanosine cap;
17. additional of 3'-poly 'A' tail;

<u>Energetics: max 1</u>

18. idea that aerobic respiration releases energy (ATP) to support the various stages of immunoglobin production;

QWC (1m) points communicated clearly <u>without ambiguity</u> and with <u>relevant connections</u> as to how haemoglobin (protein) is expressed as HSC differentiate to form RBC

[Total: 15]

7 **(a)** With reference to named examples, describe the range of roles performed by ATP in living organisms. [6]

1. ATP – adenosine triphosphate, an energy currency in the cell produced by phosphorylation of ADP + P_i ;

2. when hydrolysed into ADP + P_i, releases a lot of energy to fuel many anabolic reactions within the cell ;;

3. In cellular/aerobic respiration: Activation of glucose to glucose-6-phosphate via phosphorylation using ATP during the energy investment phase of glycolysis.

OR

- ATP is required for the phosphorylation of fructose-6-phosphate to fructose-1,6-bisphosphate catalysed by phosphofructokinase ;
- ATP also serves as an allosteric inhibitor to the phosphofructokinase enzyme ;

4 needed in the endomembrane system i.e. to supply energy needed to power the migration of vesicles (transport / secretory) between organelles for protein / lipid trafficking ;;

5 Active transport of molecules against concentration gradient across the cell surface membrane via the action of a specific carrier proteins called "pump" which use ATP to change its conformation. E.g. is sodium-potassium pump ;

6 required for active transport processes i.e. to move large substances against concentration gradient through carrier proteins / in bulk transport processes, endocytosis and exocytosis ;;

7 synthesis of large biomolecules (proteins, carbohydrates) through formation of bonds between monomers (peptide bonds, glycosidic bonds / amino acids, monosaccharides) requires energy ;;

8 ATP required for amino acid activation prior to translation for the covalent attachment of the amino acid to the 3' acceptor stem of the corresponding tRNA, catalysed by amino acyl tRNA synthetase;

In photosynthesis:

8. energy from ATP is required in the convert glycerate-3-phosphate (GP) to glyceraldehyde-3-phosphate (GALP) in Calvin cycle, as well as in the regeneration of RuBP during light independent stage ;;
9. Energy from hydrolysis of ATP is also needed to regenerate ribulose bisphosphate in Calvin cycle ;

Role of ATP as a nucleotide:

10. ATP is a ribonucleotide and is incorporated into mRNA acid during transcription ;

Others

I. phosphorylation of proteins, where a phosphate group from ATP is added to proteins → activating / inactivating proteins by triggering a 3D conformational change → e.g. kinases in cell cycle /signal transduction pathways ;;

II. polymerisation of microtubules during cell division requires ATP → MTs can then be attached to kinetochores of chromosomes to align them at the metaphase plate / elongation of the cell ;;

III. act as an allosteric inhibitor in respiration i.e. high ATP inhibits glycolysis (phosphofructokinase enzyme) / Krebs cycle ;;

IV. ATP/AMP ratio acts as a biosensor in bacteria → low levels of ATP corresponding to high levels of AMP triggers increased transcriptional rates of the *lac* operon ;;

QWC (2m) points communicated clearly <u>without ambiguity</u> and with <u>relevant examples</u> as to how ATP is useful in living organisms (plants, animals, prokaryotes)

[Total: 6]

(b) Cytochrome *c* is primarily known as an electron-carrying mitochondrial protein. The amino acid sequence is highly conserved in eukaryotes, differing by a few residues.

Cytochrome *c* amino acid sequence in humans is more similar to that of chimpanzees but differs more from that of horses.

Explain how cytochrome *c* was conserved over time in eukaryotes. [9]

[Total: 15]

1 Cytochrome c important protein as participates in electron transfer as part of the mitochondrial electron transport chain (ETC)

2 is thus an indispensable part of the energy production process ATP in oxidative phosphorylation

3 eukaryotes : all have mitochondria in their cells for respiration

4. The variations in the amino acid sequences for cytochrome c are due to mutations in ancestral populations

5. There are different selection pressures that will select for advantageous alleles that express out functional cytochrome c

6 these advantageous alleles passed on to offsprings who survive to sexual maturity and reproduce

7 Increase in frequency of alleles expressing functional cyc c accumulate in the population

8 As new species evolved by the process of natural selection ,

9 any mutation that lead to non functional cytochrome c will lead to that variant not able to survive as unable to undergo respiration

10 because 3D conformation of protein disrupted

11 Due to ionic bonds or hydrogen bonds not formed between the R groups of the amino acids

12 The humans, horses could have mutations that occurred in the different eukaryotes : silent mutations

13 Mutation that are on non essential genes

14 Cyc c still functional and conserved in eukaryotes

[Total: 9]

QWC (1m) points communicated clearly